Roll No.....

ODD MID SEMESTER EXAMINATION SEPTEMBER - 2025

Name of the Program: **M. TECH** | *Semester:* **I**

Name of the Course: **Advance Power Electronics** | *Course Code:* **MEV 102**

Time: 90 Minutes | Maximum Marks: 50

Note:

(i) Answer **all the questions** by choosing ***any one of the sub questions***.

(ii) Each question carries 10 marks

Q1	(10 marks)	CO 1
a)	What is the operating principle of power MOSFET? Describe its switching characteristics.	
	OR	
b)	Explain latching and holding current of SCR indicated in V-I characteristics.	
Q2	(10 marks)	CO 1
a)	Examine the basic structure, working and the turn ON and turn OFF processes of IGBT with its equivalent circuit.	
	OR	
b)	Explain the principle of operation, advantages, and limitations of GTO. How does it differ from a conventional SCR?	
Q3	(10 marks)	CO 1 CO 2
a)	Differentiate between single-quadrant, two-quadrant, and bi-directional switches with suitable circuit diagrams.	
	OR	
b)	A boost converter has the input voltage of 24 V and it operates at 20 kHz, when the average output voltage is 12 V, the average load current is 1.2 A, it is having L = 300 μH, C= 150 μF. Find duty cycle, ripple current through the inductor, ripple voltage and critical values of L and C ?	
Q4	(10 marks)	CO 2
a)	Explain the working of buck converter with neat waveform and derive the expression for peak-to-peak voltage across the capacitor.	
	OR	
b)	The buck converter has an input voltage of Vs = 15 V. The required average output voltage is Va = 5 V at R = 400 Ω and the peak-to-peak output ripple voltage is 10 mV. The switching frequency is 20 kHz. If the peak-to-peak ripple current of inductor is limited to 0.6 A, determine: i) the duty cycle ii) the filter inductance L, iii) the filter capacitor C.	
Q5	(10 marks)	CO 2
a)	Explain the concept of Continuous Conduction Mode (CCM) and Discontinuous Conduction Mode (DCM) in DC–DC converters. Derive the condition for boundary conduction.	
	OR	
b)	Describe the role of DC–DC converters in hybrid vehicle powertrains. Why are they essential for energy management?	